Object Detection

Object Detection
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Demonstrating the effectiveness of **Ultralytics**: Object Detection in image, video, and real-time detection using Python.

1. Enabling Optimal Motherboard Performance

1.1 Enabling MAX Power Mode

Enabling MAX Power Mode on a Jetson will ensure all CPU and GPU cores are running:

```
sudo nvpmodel -m 0
```

1.2 Enabling Jetson Clocks

Enabling Jetson Clocks will ensure all CPU and GPU cores run at maximum frequency:

```
sudo jetson_clocks
```

2. Object Prediction: Images

Using yolo11n.pt to predict images included with the Ultralytics project.

Navigate to the code folder:

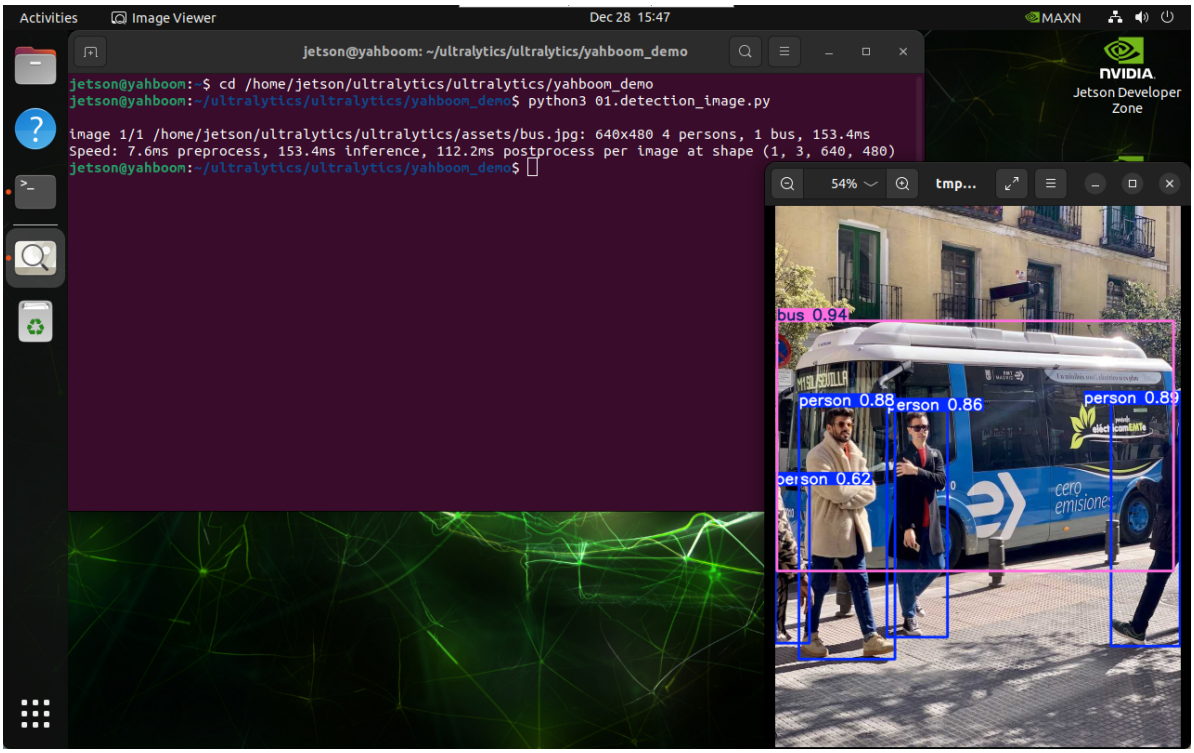
```
cd /home/jetson/ultralytics/ultralytics/yahboom_demo
```

Run the code:

```
python3 01.detection_image.py
```

Preview of the effect

Location of the image output by YOLO recognition: /home/jetson/ultralytics/ultralytics/output/



Example code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n.pt")

# Run batched inference on a list of images
results = model("/home/jetson/ultralytics/ultralytics/assets/bus.jpg")  # return
a list of Results objects

# Process results list
for result in results:
    boxes = result.boxes  # Boxes object for bounding box outputs
    # masks = result.masks  # Masks object for segmentation masks outputs
    # keypoints = result.keypoints  # Keypoints object for pose outputs
    # probs = result.probs  # Probs object for classification outputs
    # obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen

 result.save(filename="/home/jetson/ultralytics/ultralytics/output/bus_output.jp
g")  # save to disk
```

3. Target Prediction: Video

Use yolo11n.pt to predict videos in the Ultralytics project (not the videos included with Ultralytics).

Enter the code folder:

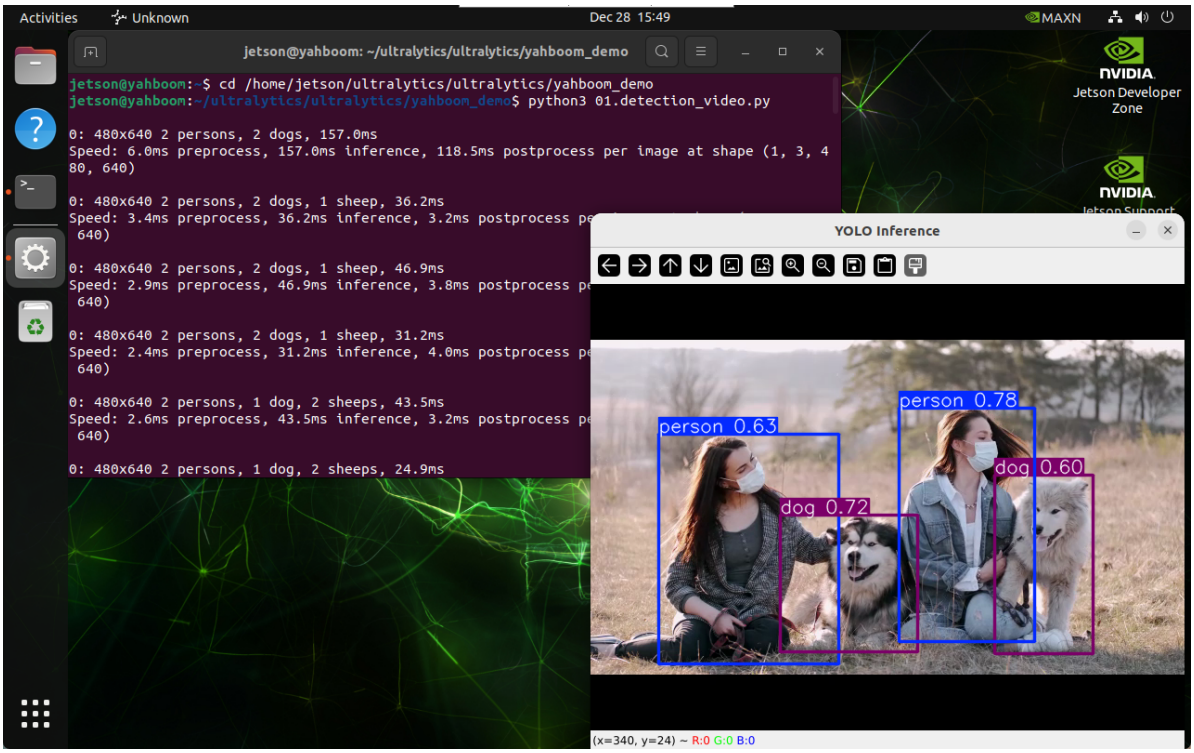
```
cd /home/jetson/ultralytics/ultralytics/yahboom_demo
```

Run the code:

```
python3 01.detection_video.py
```

Preview of Results

Location of the video output by YOLO: /home/jetson/ultralytics/ultralytics/output/



Example Code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n.pt")

# Open the video file
video_path = "/home/jetson/ultralytics/ultralytics/videos/people_animals.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path =
"/home/jetson/ultralytics/ultralytics/output/01.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()
```

```
    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", annotated_frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4. Target Prediction: Real-time Detection

4.1 USB Camera

Use yolo11n.pt to predict the image from a USB camera.

Enter the code folder:

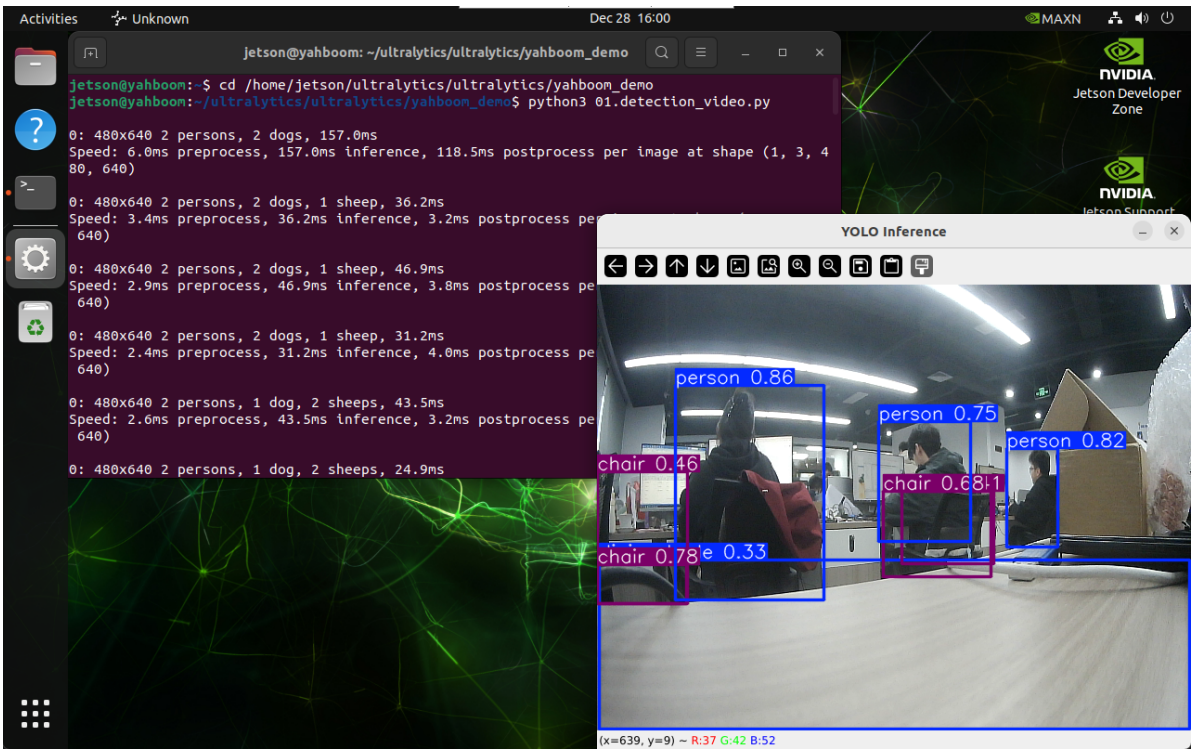
```
cd /home/jetson/ultralytics/ultralytics/yahboom_demo
```

Run the code: Click the preview screen, press q to terminate the program!

```
python3 01.detection_camera_usb.py
```

Preview of Results

Location of the video output by YOLO: /home/jetson/ultralytics/ultralytics/output/



Example Code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n.pt")

# Open the cammera
cap = cv2.VideoCapture(0)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path =
"/home/jetson/ultralytics/ultralytics/output/01.detection_camera_usb.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
```

```
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", annotated_frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

Reference Materials

https://docs.ultralytics.com/tasks/detect/